

ISP Synthetic Dataset

Data Dictionary — four tables modelled on a large Australian ISP (650k+ residential · 97k+ business · 198k+ wholesale customers)

customers.csv			~2,000 rows
Master customer registry and primary dimension table. One record per subscriber across residential, business, and wholesale segments. Captures commercial attributes (plan, contract, pricing), lifecycle dates (onboarding, churn), and add-on flags. All foreign keys in the other three tables resolve back to customer_id here. Grain: One row per unique customer Columns			
Column	Type	Notes / distribution	
■ customer_id	string	Unique customer identifier. Format: CUST-XXXXX	
customer_type	enum	Segment: residential (65%), business (20%), wholesale (15%)	
region	enum	Australian city weighted by population: Sydney, Melbourne, Brisbane, Adelaide, Perth, Canberra	
plan_type	enum	Service plan — varies by customer type. NBN50/100/250/1000 for residential; Fibre, Business_VoIP for business; Wholesale_Dark_Fibre for wholesale	
contract_type	enum	Commitment term: month-to-month, 1-year, or 2-year. Higher churn on month-to-month	
monthly_charge_aud	float	Monthly recurring charge in AUD. Plan base price +/- small random variance	
tenure_months	integer	Months since onboarding. Exponentially distributed, skewed toward longer tenure (range 1-60)	
has_cybersecurity_addon	boolean	Cybersecurity add-on active. Prevalence ~55% business, ~10% residential	
has_voip_addon	boolean	VoIP add-on active. Common for business, low for residential	
onboarded_date	date	Service activation date. Ranges from 2020-01-01 to present	
churn_flag	boolean	Whether the customer has cancelled. Overall rate ~15%; up to 25% for month-to-month contracts	
churn_date	date null	Date of cancellation. Null for active customers	

network_telemetry.csv

~44,800 rows

Daily network performance readings for a rolling 90-day window covering 600 sampled customers. Captures bandwidth consumption, latency, packet loss, and connection stability. Contains three simulated regional outage events causing visible spikes in latency and packet loss. Churning customers show degrading metrics in the 30 days before their churn_date.

Grain: One row per customer per day (90-day window, 600 customers)

Columns

Column	Type	Notes / distribution
■ telemetry_id	integer	Auto-incrementing row identifier
customer_id	string	Links to customers.csv. Sampled from 600 customers
date	date	Reading date. Last 90 days from generation date
region	enum	Denormalised from customer record for query convenience
plan_type	enum	Denormalised plan — contextualises bandwidth norms
daily_bandwidth_gb	float	Total data consumed that day. Mean varies by plan: NBN50 ~8 GB, NBN1000 ~90 GB, Dark Fibre ~400 GB
peak_hour_latency_ms	float	Average latency during peak hours. Normally distributed by region; spikes to 60+ ms during outages
off_peak_latency_ms	float	Average latency off-peak. Typically 40-70% of peak value
packet_loss_percent	float	Daily packet loss %. Near zero normally; spikes to 2.5-6% during outage events
connection_drops	integer	Number of connection interruptions per day (0-5). Higher during outages and pre-churn
sla_breach	boolean	True if peak latency >100ms OR packet loss >3%. Threshold is 80ms for business/wholesale plans
outage_flag	boolean	True on days matching one of 3 simulated regional outage windows
node_id	string	Network node serving this customer. Format: NODE-XX. ~20 nodes across all regions

support_tickets.csv

~3,000 rows

Customer support interactions over a 180-day window. Each row is one ticket from open to close. Ticket volume spikes around the three outage events in affected regions — a correlation detectable via date-range joins with the telemetry table. Resolution time drives satisfaction scores, with faster resolution correlating with higher ratings.

Grain: One row per support ticket

Columns

Column	Type	Notes / distribution
■ ticket_id	string	Unique ticket identifier. Format: TKT-XXXXX
customer_id	string	Links to customers.csv
open_date	date	Date ticket was raised. Within last 180 days; biased toward outage windows for affected regions
close_date	date null	Date ticket was resolved (open_date + 1-14 days). Null if still open
issue_category	enum	connectivity_drop (30%), slow_speed (25%), billing_query (20%), hardware_fault (10%), outage_report (10%), provisioning_delay (5%)
region	enum	Customer region — used for outage correlation analysis
channel	enum	Contact channel: phone (35%), email (25%), chat (20%), web_portal (20%)
resolved	boolean	Whether the issue was resolved. ~82% resolution rate overall
satisfaction_score	int null	Post-resolution CSAT score (1-5). Null if unresolved. Higher scores for tickets closed within 2 days

billing_events.csv

~20,500 rows

Monthly billing records for all customers across a 12-month window. Each row represents one invoice cycle. Payment method is consistent per customer. Billing stops after a customer's churn_date, so churned customers have fewer records. Late and failed payments can identify at-risk accounts or be correlated with eventual churn.

Grain: One row per customer per billing month

Columns

Column	Type	Notes / distribution
■ billing_id	integer	Auto-incrementing row identifier
customer_id	string	Links to customers.csv
billing_month	date	First day of the billing month (e.g. 2024-11-01). Covers last 12 calendar months
amount_charged_aud	float	Invoice amount in AUD. Customer base monthly_charge +/- small random variance (credits, one-off fees)
payment_status	enum	Outcome: paid (85%), late (10%), failed (5%)
payment_date	date null	Date payment received. 1-7 days after billing_month if paid; 8-30 days if late; null if failed
payment_method	enum	Consistent per customer: direct_debit (55%), credit_card (30%), bank_transfer (15%)